

Homework Classical Physics, Fall 2023

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Problem Set 02.4 (L03 & L04)

Due on Thursday, 21. September 2023, at 09:30 (in the morning!)

(= Deadline for submitting a pdf file with solution)

Mechanics - Projectile motion (4 points)

A massive ball is falling down from an initial height of $h_{\text{ball}} = 20 \text{ m}$. With a gun held horizontally, at a distance $d = 50 \text{ m}$ from the trajectory of the falling ball, at a height of $h_{\text{gun}} = 10 \text{ m}$, we are going to shoot at the falling ball. The bullet leaves the gun at a speed of $v = 100 \text{ m/s}$.

At what time after the start of the fall should the gun be fired in order to hit the falling ball with the bullet? (neglect air resistance)